Engine Model(s): PT6A

73-10-07

PNEUMATIC LINE (P3) - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC03-001	Oil, Engine Lubricating
PWC05-007	Fluid, Check
PWC05-031	Flux, Resin
PWC05-052	Sheathing, Heat-Shrinkable
PWC08-002	Adhesive, Fluorosilicone
PWC09-001	Sealant, Silicone Rubber
PWC11-002	(Use PWC11-014)
PWC11-014	Alcohol, Isopropyl
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

The fixtures, equipment and supplier tools listed below are used in the following procedures.

Name	Remarks
Cleaner, Electrosonic	

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Name	Remarks
Turbine Temperature Indicating System Test Set	TT1000A or TT1200A

5. Servicing

A. Procedure

(1) Servicing of the pneumatic system consists of cleaning or replacement of the disposable air filter element.

(2) Replacement of element is in accordance with Chapter 72-00-00, Table 601 or 602.

(3) Indications of need to change the element are no, or slow, response to power control lever movements during engine run, when it is known there are no leaks in the pneumatic system.

(4) It is recommended, therefore, that operators monitor engine response characteristics and derive their own cleaning/replacement schedules.

6. Removal/Installation

CAUTION: HOLD THE FITTING CORRECTLY IN POSITION WHEN YOU TORQUE THE COUPLING NUT.

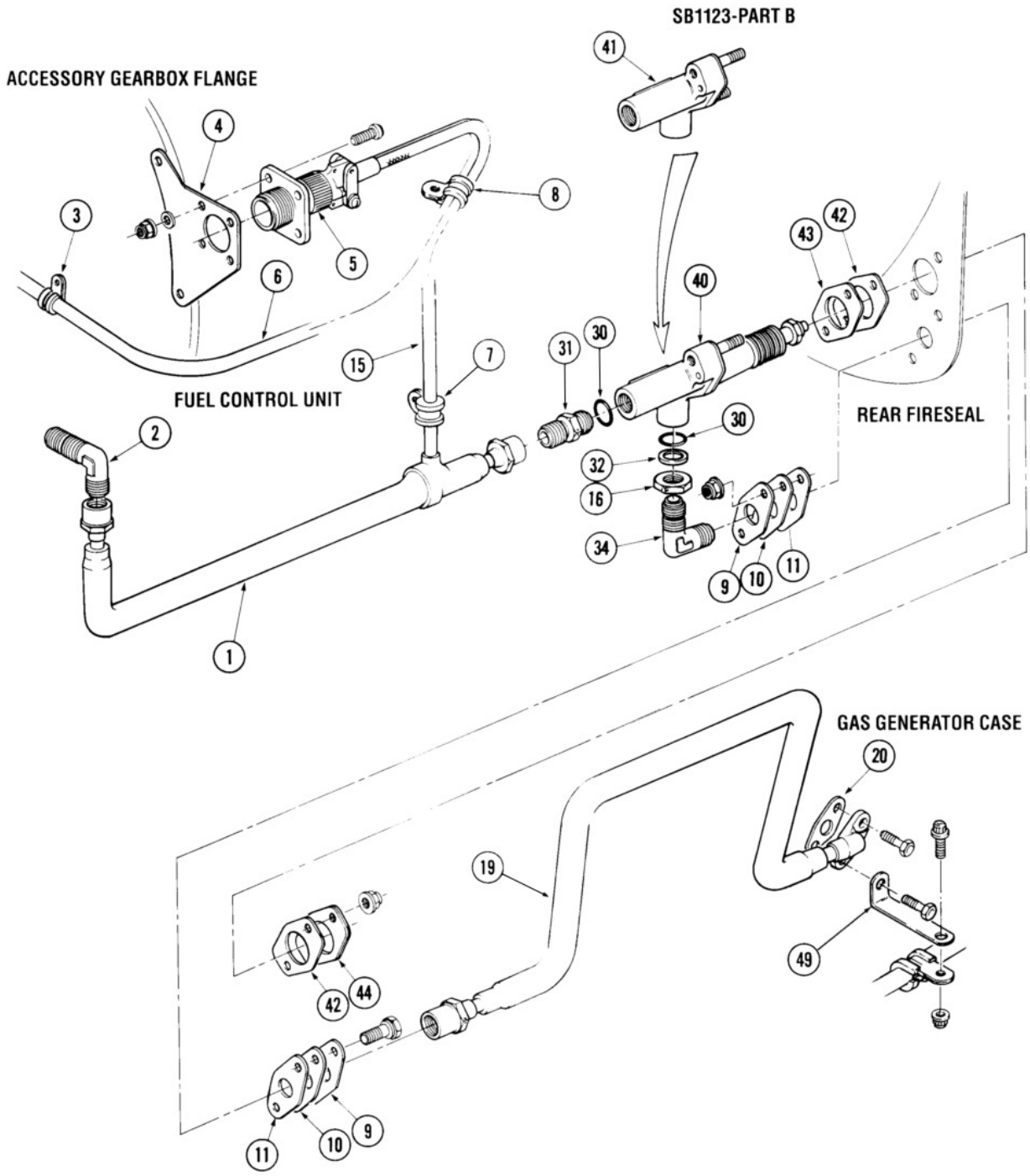
NOTE: It is important to make sure that all solid P3 lines are properly and correctly fitted before the coupling nuts are tightened. The coupling nuts should be seated without the use of tools (finger tight) before torquing. The installation of any pipe brackets should not distort the pipe in any way.

A. Removal of Air Pressure Lines (Ref. Fig. 201)

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Figure 201 Removal/Installation of P3 Air Pressure Lines

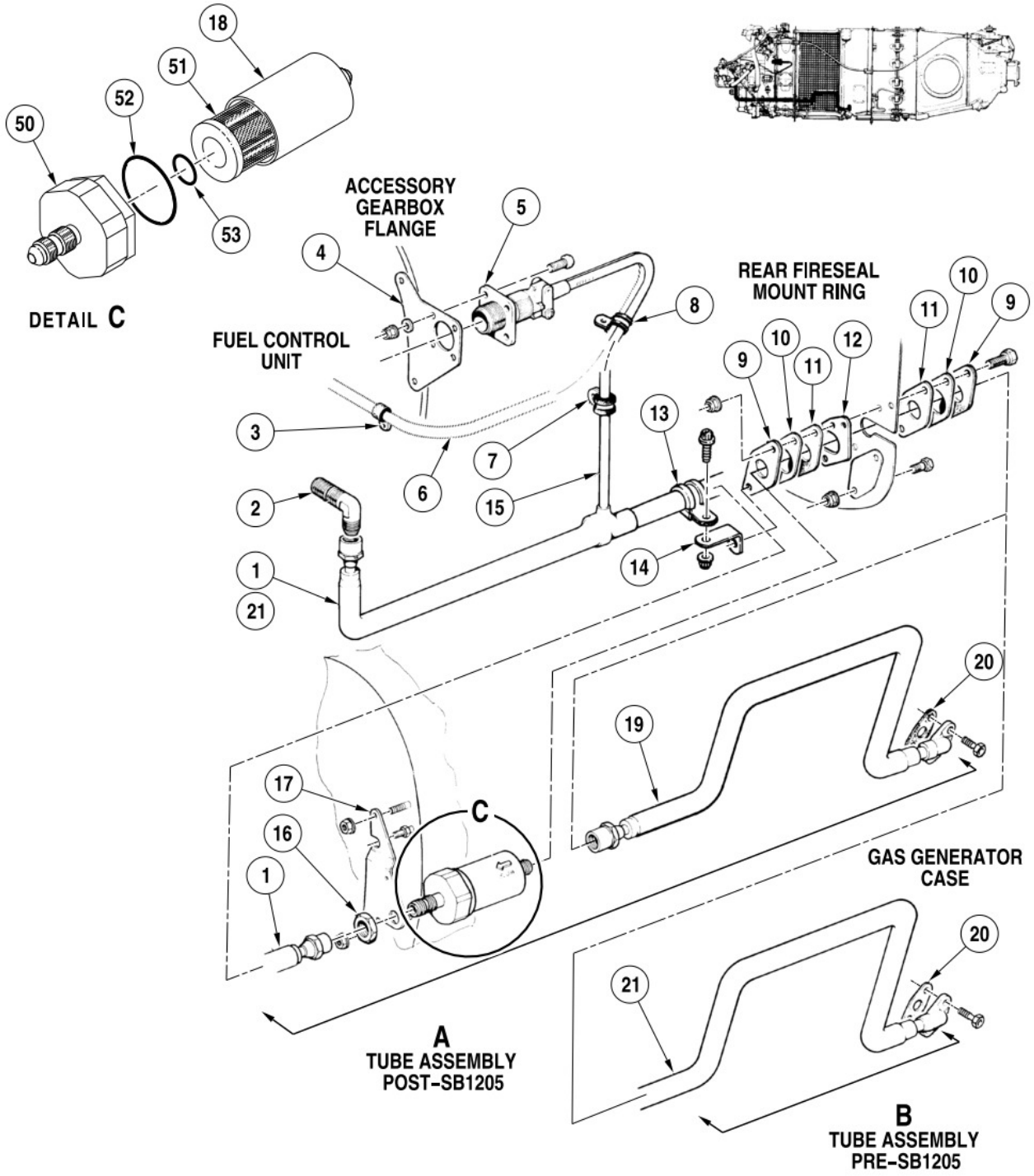
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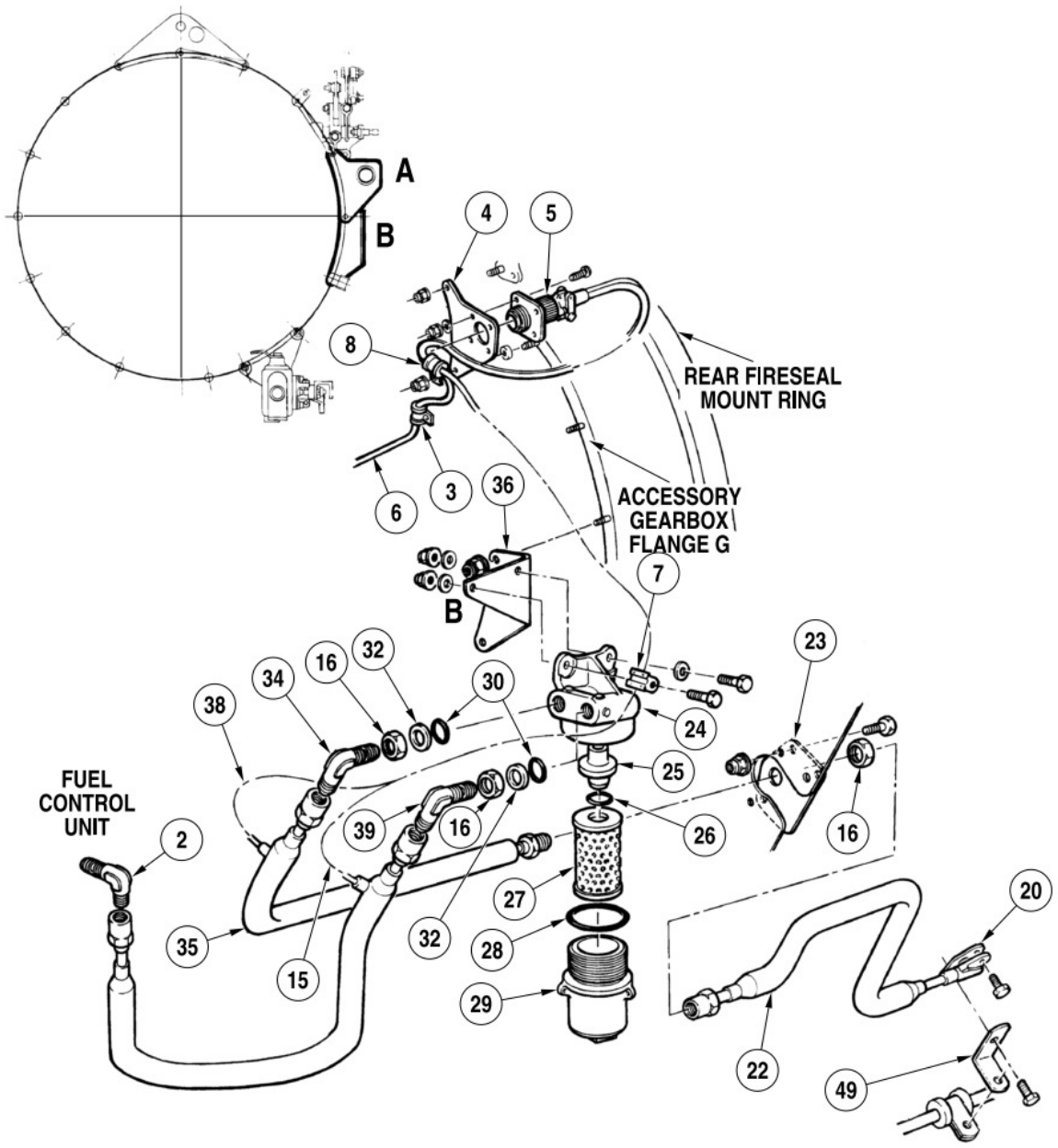
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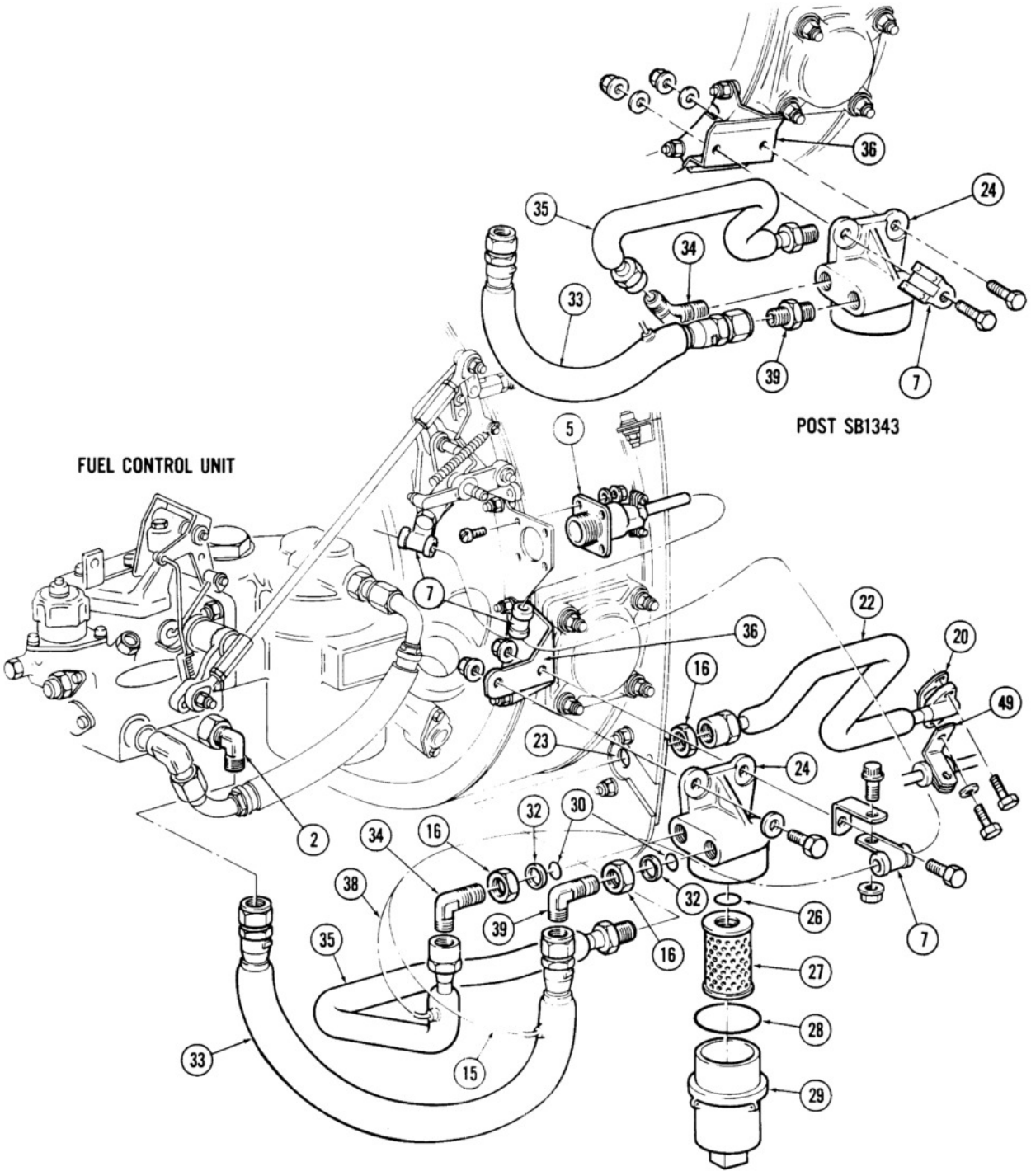


Post-SB1378

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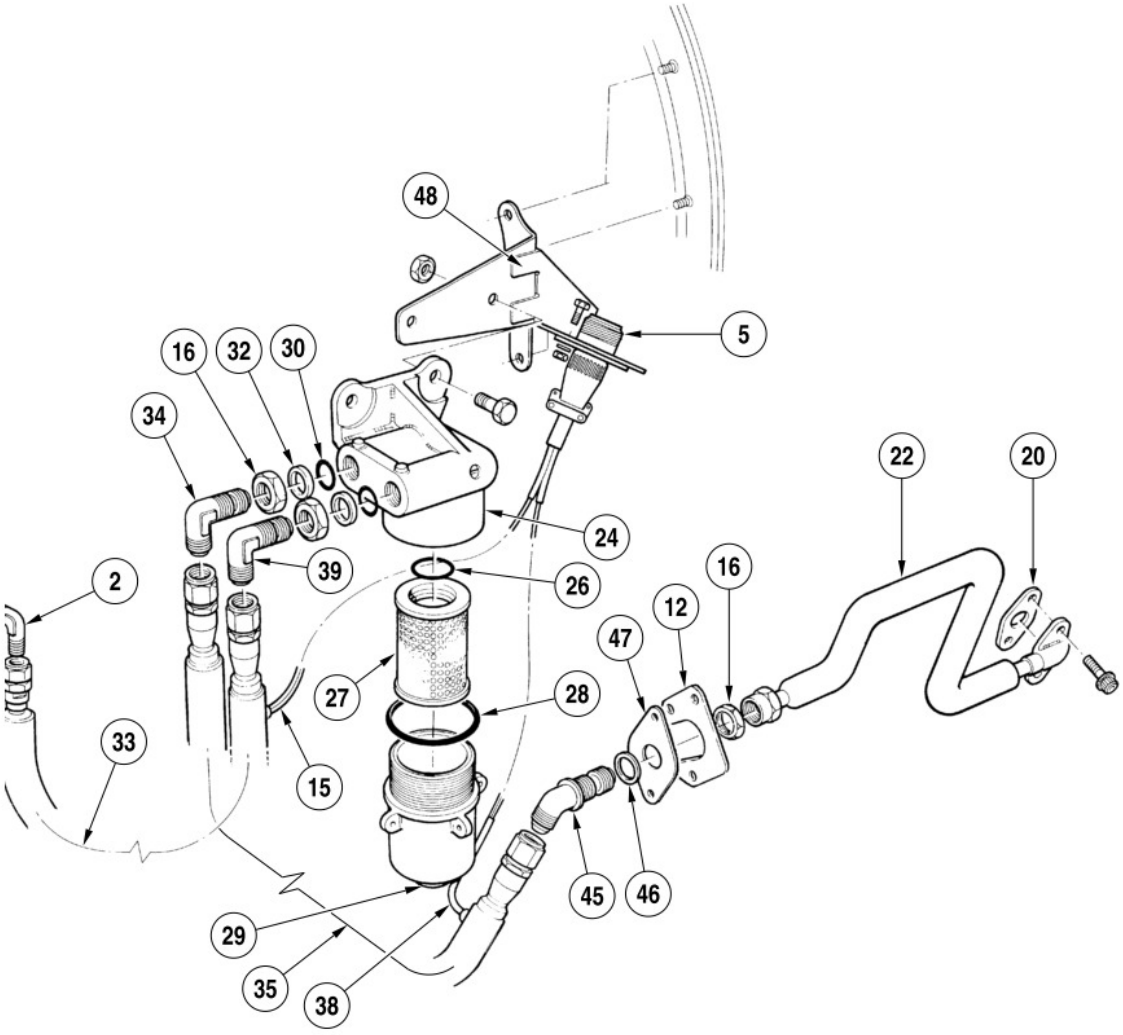


Post-SB1330 and Post-SB1343

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(SHEET 5 OF 5)



Post-SB1448

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1. Heated Air Pressure Line (P3 Rear)
2. Metered Elbow (on FCU)(Except unmetered Pre-SB1123)
3. Clamp
4. Support Bracket
5. Electrical Connector
6. Electrical Cable (Py Line)
7. Clamp
8. Clamp
9. Retaining Plate
10. Insulation
11. Seal
12. Slotted Retaining Plate
13. Clamp
14. Bracket
15. Electrical Cable (P3 Line)
16. Locknut
17. Support Bracket
18. Air Filter Cover (Post-SB1205)
19. Insulated Air Pressure Line (P3 Front)(Post-SB1205)
20. Gasket
21. Heated Air Pressure Line (P3)(Pre-SB1205)
22. Insulated Air Pressure Line (P3 Front)
23. Support Plate
24. Air Filter Housing
25. Transfer Tube (Ref.)
26. Preformed Packing
27. Element (3-micron)
28. Preformed Packing
29. Air Filter Cover
30. Preformed Packing
31. Straight Nipple
32. Back-up Ring
33. Heated Air Pressure Hose (P3 Rear)(Post-SB1330/Post-SB1343/Post-SB1448)
34. Elbow
35. Heated Air Pressure Tube (P3 Center)(Post-SB1330/Post-SB1378)
Heated Air Pressure Hose (Post-SB1343/Post-SB1448)
36. Support Bracket
37. Not Used
38. Electrical Cable (P3 Line Center)
39. Elbow
40. Temperature Compensator (Pre-SB1123)
41. Compensator Body (Post-SB1123, Part B)
42. Seal
43. Insulation
44. Retaning Plate and Insulation Assembly
45. Elbow Tube (Post-SB1448)

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46. Washer (Post-SB1448)
47. Retaining Plate (Post-SB1448)
48. Support Bracket Assembly (Post-SB1448)
49. Bracket (Post-SB1196/Post-SB1448)
50. Hexagon Head
51. Filter Element
52. Preformed Packing
53. Preformed Packing

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(1) Disconnect electrical cable (15 or 38) of relevant heated line (1, 21, 33 or 35) from electrical connector (5):

(a) Detach and remove clamps (7 and 8) from their respective locations.

(b) Disconnect cables (15 and 38) from connector (Ref. Subpara. D.).

(2) For PT6A-27 and PT6A-28 Engines: Remove air pressure lines (1 and 19):

(a) Disconnect and remove heated rear line (1) from elbow (2) on FCU and from straight nipple (31) on temperature compensator (40) or compensator body (41).

(b) Loosen coupling nut of insulated front line (19) at elbow (34) on temperature compensator or compensator body.

(c) Remove two bolts and bracket (49, Post-SB1196) if fitted, securing flanged elbow of front line (19) to boss on gas generator case. Discard gasket (20).

(d) Remove insulated front line (19) by disconnecting from elbow (34).

(3) For PT6A-27 and PT6A-28 Pre-SB1205 Engines and all PT6A-21 Engines: Remove heated air pressure line (21):

(a) Detach and remove clamp (13) from bracket (14) by removing self-locking nut and bolt.

(b) Remove two self-locking nuts, bolts and sealing components (9, 10, 11 and 12) from rear fireseal mount ring.

(c) Remove bolts and bracket (49, Post-SB1196) if fitted, securing flanged elbow of line (21) to boss on gas generator case. Discard gasket (20).

(d) Disconnect line from metered elbow (2) on FCU. Remove line complete with sealing components from slot in rear fireseal.

(e) Remove sealing components (9, 10 and 11) from line.

(4) For PT6A-27 and PT6A-28 Post-SB1205 Engines:

(a) Disconnect and remove heated rear line (1) from metered elbow (2) on FCU and from fitting on body of air filter (18).

(b) Loosen coupling nut of insulated front line (19).

(c) Remove bolts securing flanged elbow and bracket (Post-SB1196) if fitted, to boss on gas generator case. Discard gasket (20).

(d) Remove insulated front line by disconnecting from fitting on air filter (18).

(5) For PT6A-28 Post-SB1290, PT6A-21/-27/-28 Post-SB1330/Post-SB1343/Post-SB1378 Engines:

(a) Disconnect and remove heated rear line (33) from metered elbow (2) on FCU and elbow (39) on air filter housing (24).

NOTE: To improve resistance to cracking or fracture, theheated and insulated compressor delivery tubes (33 and 35) (Post-SB1290, Pre-SB1314) are removed

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and replaced (Post-SB1314) with tubes having increased wall thickness.

(b) Loosen coupling nut of insulated front line (22).

(c) Remove two bolts securing flanged elbow of line (22) and bracket (49,Post-SB1196) if fitted, to boss on gas generator case. Discard gasket (20).

(d) Remove front line (22) by disconnecting from center line (35) at rear fireseal.

(e) Remove locknut (16) securing fitting of heated center line (35) to support plate (23) at rear fireseal.

(f) Remove line (35) by disconnecting from elbow (34) on air filter housing (24) and withdrawing from plate (23).

(6) For PT6A-27 and PT6A-28 Post-SB1448 Engines:

(a) Disconnect coupling nuts of heated rear hose (33) from elbow (2) in FCU and elbow (39) in filter housing (24).

(b) Disconnect coupling nuts of heated center hose (33) from elbow (34) in filter housing (24) and elbow tube (45) through rear fireseal mount ring.

(c) With electrical cables disconnected (Ref. Subpara. B.(1)) remove heated center and rear hoses (33 and 35).

(d) Loosen insulated tube (22) coupling nut at rear fireseal mount ring, and remove bolts and flanged elbow. Disconnect coupling nut and remove insulated tube. Discard gasket (20).

(e) If necessary to remove elbow (45), remove locknut (16) and remove elbow rearward through rear fireseal mount ring. Remove and retain washer (46) from elbow tube.

B. Removal of Air Filter (Ref. Fig. 201)

(1) For PT6A-27 and PT6A-28 Post-SB1205 Engines:

(a) Remove locknut (16) and remove air filter (18) from support bracket (17) at flange G of accessory gearbox.

(b) Remove hexagon head (50) from housing of air filter cover (18) and remove filter element (51). Discard preformed packing (52) from groove in hexagon head and small preformed packing (53) from inner, flanged diameter of element (51).

(2) For PT6A-28 Post-SB1290, and PT6A-21/-27/-28 Post-SB1330/Post-SB1343/Post-SB1378 Engines:

CAUTION: USE EXTREME CARE WHEN P3 FILTER COVER IS REMOVED IN ORDER TO AVOID DAMAGE TO FCU/START CONTROL TELESCOPIC ROD. DAMAGE TO THIS ROD CAN CAUSE HANG-UP AT HIGH POWERS. EXCESS TORQUE OR UNTORQUING LOADS CAN BEND FILTER BRACKET, THUS CAUSING INTERFERENCE WITH CONTROL ROD.

(a) Remove assembly of air filter from support bracket (36) at flange G, by removing two bolts and clamp (7).

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NOTE: Step (a) may be omitted if filter element only isbeing serviced.

(b) Remove cover (29) from housing (24). Discard preformed packing (28).

(c) Remove filter element (27).

NOTE: Transfer tube (25, Post-SB1290) should remain in-situ.

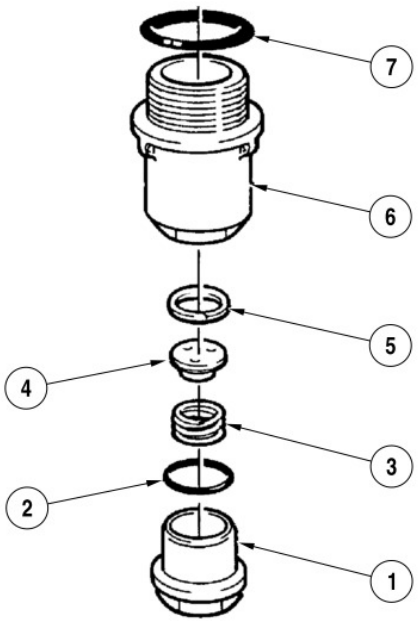
(d) Remove and discard preformed packing (26) from inner, flanged diameter of element.

(e) If a new filter housing is to be assembled at installation, remove elbow (34), elbow (39) and locknuts (16). Discard preformed packings (30) and back-up rings (32).

NOTE: For Post-SB1495 Engines: Installation of air filter drain valve (Ref. Fig. 202), complete steps (f) through (h) as follows in addition to (a) through (e) above:

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Figure 202 Removal/Installation of Air Filter Housing Drain Valve



Post-SB1495

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1. Drain Valve Adapter
2. Preformed Packing
3. Spring
4. Compressor Air Drain Valve
5. Retaining Ring
6. Filter Cover
7. Preformed Packing

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(f) Remove filter cover (6), and discard preformed packing (7).

(g) Remove drain valve adapter (1), and discard preformed packing (2).

(h) Remove retaining ring (5), drain valve (4), and spring (3).

(3) For PT6A-27 and PT6A-28 Post-SB1448 Engines:

(a) Remove filter cover (29) from housing assembly (24). Discard preformed packing (28).

(b) Remove filter element (27) from housing assembly (24). Discard preformed packing (26) from inner flanged diameter of filter element.

(c) If new filter housing is to be installed, loosen locknuts (16) on elbows (34 and 39) and remove elbows from filter housing (24). Remove preformed packings (30), back-up ring (32) and locknuts from elbows. Discard preformed packings.

(d) Remove bolts and filter housing (24) from bracket (48).

NOTE: For Post-SB1495 Engines: Installation of air filter drain valve (Ref. Fig. 202), complete steps (e) through (g) as follows in addition to (a) through (d) above:

(e) Remove filter cover (6), and discard preformed packing (7).

(f) Remove drain valve adapter (1), and discard preformed packing (2).

(g) Remove retaining ring (5), drain valve (4), and spring (3).

C. Removal of Temperature Compensator (PT6A-27 and PT6A-28) (Ref. Fig. 201)

(1) Disconnect heated line (1) from straight nipple (31) on compensator.

(2) Disconnect insulated line (19) from elbow (34) on compensator.

(3) Remove two self-locking nuts, compensator (40 or 41), seals (42), insulation (43) and retaining plate assembly (44) from rear fireseal mount ring.

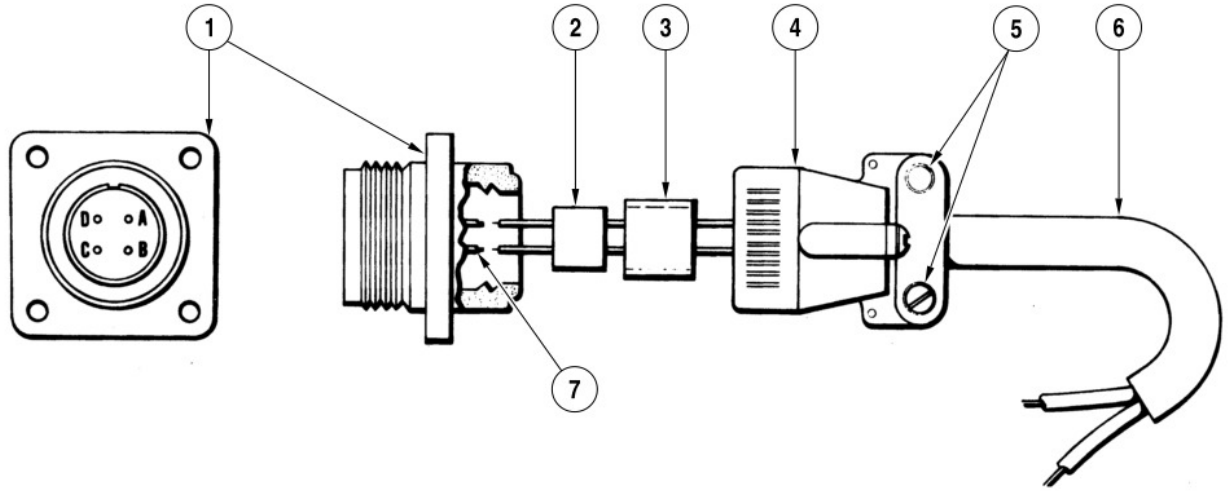
(4) If replacement compensator is to be fitted, remove straight nipple (31) and elbow (34) from compensator. Discard preformed packings (30).

(5) Remove self-locking nuts, bolts, seals (11), insulation (10) and retaining plates (9) from rear fireseal mount ring, if insulation or seals are damaged.

D. Removal of Electrical Connection of Heated Lines (Ref. Fig. 203)

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Figure 203 Heated Lines - Wiring Details



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1. Connector Shell
2. Rubber Insulator
3. Nylon Sleeve
4. Bell End Fitting
5. Clamp Screws
6. Cable Sleeve
7. Terminal Pins

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(1) Disconnect leads of cable, of relevant heated air pressure line from connector:

(a) Remove two clamp screws (5) and remove clamp halves from bell end fitting (4).

CAUTION: MAKE SURE LEADS OF CABLES DO NOT TURN WHEN UNSCREWING BELL END FITTING FROM SHELL.

(b) Remove bell end fitting (4) from connector shell (1) and slide fitting and cable sleeve (6) back along cables.

(c) Slide nylon sleeve (3) and rubber insulator (2) back on cables to expose terminal pins (7).

(d) Using a suitable soldering iron, disconnect leads of compressor discharge air pressure line(s) from pins B and D; disconnect leads of propeller governor air pressure line from pins A and C.

(e) If all cables are being disconnected, remove connector components from electrical cables.

(2) Connect leads of cable, of relevant heated air pressure line to connector:

(a) If all cables were disconnected, reassemble connector components on cables:

1 Insert cables through sleeve (6), bell end fitting (4) and nylon sleeve (3). Slide components back along cables.

2 Pass each lead of cable(s) through relevant holes in rubber insulator (2).

(b) Using a suitable soldering iron, with solder (PWC05-205) and resin flux (PWC05-031), solder leads of compressor discharge air pressure line(s) to pins B and D; solder leads of propeller governor air pressure line to pins A and C.

CAUTION: DO NOT ALLOW LEADS TO TURN WHEN TIGHTENING BELL END FITTING.

(c) Slide rubber insulator (2) and nylon sleeve (3) over soldered connections. Slide bell end fitting (4) with sleeve (6), over sleeve (3) and attach to connector; screw on fingertight.

(d) Attach cables and sleeve (6) with cable clamp; attach clamp halves with screws (5).

(e) Do conductivity and resistance checks (Ref. Inspection/Check).

E. Installation of Air Pressure Lines (Ref. Fig. 201)

(1) For PT6A-27 and PT6A-28 Engines: Install air pressure lines (1) and (19):

(a) Install heated line (1) by connecting to elbow (2) on FCU and straight nipple (31) on temperature compensator (40) or compensator body (41). Torque coupling nuts 90 to 100 lb.in.

NOTE: Elbow (2) is metered Post-SB1123, and unmetered Pre-SB1123.

(b) Loosely connect coupling nut of insulated line (19) to elbow (34) on temperature compensator, attach flanged elbow of line and gasket (20) to boss on gas generator case with bolts. Do not tighten bolts.

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NOTE: Fit bracket (49, Post-SB1196) if required.

(c) Coupling nut of insulated line (19) is to be torqued and lockwired after pressure check (Ref. Para. Adjustment/Test).

(d) Connect leads of electrical cable (15) to relevant pins of connector (5).

(e) Install electrical cable (15) to stud at flange G with clamps (7) and self-locking nut. Torque nut 36 to 40 lb.in.

(f) Using self-locking nut, attach cables (15 and 6) together with clamp (8) to stud at lower mounting of support bracket (4). Torque 36 to 40 lb.in.

(2) For PT6A-27 and PT6A-28 Pre-SB1205 and PT6A-21 Engines:

(a) Sequentially install retaining plates (9), insulation (10) and seals (11) on heated line (21); locate line as shown in slot in rear fireseal mount ring.

(b) Install flanged elbow of line (21), gasket (20) and bracket (49) if fitted, on gas generator case with bolts. Do not tighten bolts.

(c) Connect heated line to metered elbow (2) on FCU. Coupling nut is to be fingertight only. Nut must be torqued and lockwired after pressure check (Ref. Para. Adjustment/Test).

(d) Attach line to bracket (14) with clamp (13), bolt and self-locking nut. Torque bolt 36 to 40 lb.in.

(e) Install slotted plate (12) on line and fasten sealing components (9, 10 and 11) together with slotted plate to rear fireseal mount ring with bolts and self-locking nuts. Tighten nuts until firm contact is obtained, plus 180 degrees.

(f) Connect leads of electrical cable (15) to relevant pins of connector (5).

(3) For PT6A-27 and PT6A-28 Post-SB1205 Engines: Install air pressure lines (1 and 19):

(a) Install heated line (1) by connecting to metered elbow (2) on FCU and fitting on air filter (18) at bracket (17). Torque coupling nuts 90 to 100 lb.in.

(b) Connect leads of electrical cable (15) and secure cable.

(c) Loosely connect coupling nut of insulated line (19) to fitting on air filter (18). Attach flanged elbow of line,gasket (20) and bracket (49) if fitted, to boss on gas generator case with two bolts. Do not tighten bolts.

(d) Coupling nut of insulated line (19) is to be torqued and lockwired after pressure check (Ref. Para. Adjustment/Test).

(4) For PT6A-28 Post-SB1290, PT6A-21/-27/-28 Post-SB1330 Engines: Install air pressure lines (33, 35 and 22):

(a) Install heated line (33) by connecting to metered elbow (2) on FCU and elbow (39) on filter housing (24). Torque coupling nuts 90 to 100 lb.in.

NOTE: To improve resistance to cracking or fracture, theheated and insulated compressor delivery tubes (33 and 35) (Post-SB1290, Pre-SB1314) are replaced (Post-SB1314) by tubes having increased wall thickness.

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(b) Install heated center line (35) by inserting front end through support plate (23) at rear fireseal mount ring and connecting rear end to elbow (34) on filter housing (24). Final torque locknut (16) 38 to 42 lb.in., and torque coupling nut 90 to 100 lb.in. Secure fitting at front end of line to plate (23) with locknut (16). Torque locknut 38 to 42 lb.in.

(c) Connect leads of cables (15 and 38) and secure cables. Secure cables at air filter with clamp (7). Tighten nut and torque 36 to 40 lb.in.

(d) Loosely connect coupling nut of insulated front line (22) to center line (35) at support plate (23); secure flanged elbow of line, gasket (20) and bracket if fitted, to boss on gas generator case with two bolts. Do not tighten bolts.

(e) Coupling nut of insulated line (22) is to be torqued and lockwired after pressure check (Ref. Para. Adjustment/Test).

(5) For PT6A-27 and PT6A-28 Post-SB1448 Engines:

(a) If applicable, install washer (46) on elbow (45) and insert elbow forward through retaining plates (47 and 12) mounted on rear fireseal mount ring and fasten in position with locknut (16) torqued 38 to 42 lb.in.

(b) Install heated center hose (35), with electrical cable (38) forward, by connecting coupling nuts to inside elbow (34) on filter housing (24) and elbow (45) through rear fireseal mount ring. Torque coupling nuts 90 to 100 lb.in.

(c) Install heated rear hose (35), with electrical cable (15) forward, by connecting coupling nuts to outside elbow (34) on filter housing (24) and elbow (2) on FCU. Torque coupling nuts 90 to 100 lb.in.

(d) Connect electrical cables (15 and 38) to connector (5) mounted on support bracket (48).

(e) Install insulated tube (22) by connecting coupling nut toelbow (45) through rear fireseal mount ring and with gasket (20) in place on gas generator case boss, attach flanged end of insulated tube and bracket, if fitted, with bolts through bracket and gasket into case boss.

NOTE: Coupling nut and flange bolts on insulated tube (22) will be torqued and lockwired after pressure check (Ref. Adjustment/Test).

F. Installation of Air Filter (Ref. Fig. 201)

(1) For PT6A-27 and PT6A-28 Post-SB1205/Pre-SB1290 Engines:

(a) If element was removed for inspection purposes, install small preformed packing (53) in flanged inner diameter of existing or new element.

(b) Install large preformed packing (52) in groove in hexagon head (50).

(c) Insert filter element (51) into air filter cover (18) and press until fully bottomed.

(d) Install air filter cover (18) on hexagon head (50), screw on until metal-to-metal contact is obtained.

(e) Install assembly of air filter cover (18) on support bracket (17) at flange G with locknut (16). Torque locknut 38 to 42 lb.in.

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(2) For PT6A-28 Post-SB1290, and PT6A-21/-27/-28 Post-SB1330/Post-SB1343/Post-SB1378 Engines:

CAUTION: USE EXTREME CARE WHEN P3 FILTER COVER IS INSTALLED IN ORDER TO AVOID DAMAGE TO FCU/START CONTROL TELESCOPIC ROD. DAMAGE TO THIS ROD CAN CAUSE HANG-UP AT HIGH POWERS. EXCESS TORQUE OR UNTORQUING LOADS CAN BEND FILTER BRACKET, THUS CAUSING INTERFERENCE WITH CONTROL ROD.

(a) Install spring (3) and drain valve (4) into adapter (1); install retaining ring (5). Make sure that valve moves smoothly and freely and that retaining ring is installed properly.

NOTE: 1. For Pre-SB1495 Engines: Installation of air filter drain valve (Ref. Fig. 202), omits steps (a) through (c) and commence assembly of air filter assembly with step (d).

NOTE: 2. For Post-SB1495 Engines: Installation of air filter drain valve (Ref. Fig. 202), complete steps (a) through (c) in the sequence shown and then do (d) through (i).

CAUTION: THE P3 FILTER COVER IS MADE FROM ALUMINUM ALLOY. TAKE CARE NOT TO DAMAGE THE COVER ASSEMBLY WHEN LOCKWIRING THE DRAIN VALVE ASSEMBLY TO THE COVER ASSEMBLY. BEFORE LOCKWIRING, EXAMINE THE LOCKWIRE ATTACHMENT HOLES ON THE COVER ASSEMBLY FOR EVIDENCE OF TEARING. IF TEARING IS EVIDENT, SECURE LOCKWIRE TO AN ALTERNATE SERVICEABLE LOCATION (REF. 70-00-00, STANDARD PRACTICES - MAINTENANCE PRACTICES).

(b) Install preformed packing (2) on adapter (1), and install valve assembly in cover. Tighten 225 to 250 lb.in., and lockwire adapter to cover.

(c) Install preformed packing (7) on filter cover (6). Lubricate threads of cover with oil, and install in filter housing. Tighten 60 to 70 lb.in., and lockwire.

CAUTION: IT IS IMPORTANT THAT THE INSTALLATION OF THE DRAIN VALVE BE IN THE EXACT SEQUENCE PROVIDED ABOVE. FAILURE TO OBSERVE THIS SEQUENCE WILL RESULT IN OVERTORQUE OF THE P3 FILTER CASE.

(d) If elbows (34 and 39) were removed from filter housing (24), reinstall (Ref. 70-00-00, REMOVAL/INSTALLATION). Do not final torque locknuts (16) until angle of elbow has been established.

(e) Install preformed packing (26) in flanged inner diameter of existing or new filter element (27).

(f) Install preformed packing (28) on grooved shoulder of cover (29).

(g) Install filter element (27) and preformed packing (26) over transfer tube (25, Post-SB1290) and press in until fully bottomed in housing (24).

(h) Apply engine oil (PWC03-001) on threads of cover (29) and install on housing (24). Torque

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60 to 70 lb.in. and lockwire.

NOTE: Step (i) following may be omitted if filter element only has been serviced.

(i) Install assembly of air filter on support bracket (36) and attach with bolts, clamp (7), washers and self-locking nuts. Tighten nut at right-hand mounting and torque 36 to 40 lb.in., left-hand nut to be fingertight as clamp (7) is used to attach electrical cable of heated line at a later stage of assembly.

(3) For PT6A-27 and PT6A-28 Post-SB1448 Engines:

NOTE: 1. For Pre-SB1495 Engines: Installation of air filter drain valve (Ref. Fig. 202), omit steps (a) through (c) and commence assembly of air filter assembly with step (d).

NOTE: 2. For Post-SB1495 Engines: Installation of air filter drain valve (Ref. Fig. 202), complete steps (a) through (c) in the sequence shown and then do (d) through (g).

(a) Install spring (3) and drain valve (4) into adapter (1); install retaining ring (5). Make sure that valve moves smoothly and freely and that retaining ring is installed properly.

CAUTION: THE P3 FILTER COVER IS MADE FROM ALUMINUM ALLOY. TAKE CARE NOT TO DAMAGE THE COVER ASSEMBLY WHEN LOCKWIRING THE DRAIN VALVE ASSEMBLY TO THE COVER ASSEMBLY. BEFORE LOCKWIRING, EXAMINE THE LOCKWIRE ATTACHMENT HOLES ON THE COVER ASSEMBLY FOR EVIDENCE OF TEARING. IF TEARING IS EVIDENT, SECURE LOCKWIRE TO AN ALTERNATE SERVICEABLE LOCATION (REF. 70-00-00, STANDARD PRACTICES - MAINTENANCE PRACTICES).

(b) Install preformed packing (2) on adapter (1), and install valve assembly in cover. Tighten 225 to 250 lb.in., and lockwire adapter to cover.

(c) Install preformed packing (7) on filter cover (6). Lubricate threads of cover with oil (SB1001), and install in filter housing. Tighten 60 to 70 lb.in., and lockwire.

CAUTION: IT IS IMPORTANT THAT THE INSTALLATION OF THE DRAIN VALVE BE IN THE EXACT SEQUENCE PROVIDED ABOVE. FAILURE TO OBSERVE THIS SEQUENCE WILL RESULT IN OVERTORQUE OF THE P3 FILTER CASE.

(d) If applicable install locknuts (16), new back-up rings (32) and new lubricated preformed packings on elbows (34 and 39). Install elbows in filter housing (24), do not torque locknuts until elbow angle is established.

(e) Install new lubricated preformed packing (26) in flanged inner diameter of cleaned or new filter element (27). Insert element over center tube in filter housing (24)and press by hand until fully bottomed in housing.

(f) Install new lubricated preformed packing (28) in grooved shoulder of cover (29) and apply engine oil to threads on cover. Install cover in filter housing (24), torque 60 to 70 lb.in. and lockwire.

NOTE: Step (vii) following may be omitted if filterelement only has been serviced.

(g) If applicable install filter housing assembly (24) on bracket (48) and attach with bolts and

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self-locking nuts torqued 36 to 40 lb.in.

G. Installation of Temperature Compensator (PT6A-27 and PT6A-28) (Ref. Fig. 201)

(1) If replacement compensator is to be installed, install straight nipple (31) and elbow (34) (Ref. 70-00-00). Torque locknut on elbow 38 to 42 lb.in. and lockwire.

(2) Install seals (11), insulation (10) and retaining plates (9), on rear fireseal mount ring and fasten with bolts and self-locking nuts. Bolt heads must be on air inlet side of fireseal.

(3) Install insulation (43) and seal (42) on studs at front end of compensator. Locate compensator on rear face of rear fireseal mount ring, install seal (42) and retaining plate and insulation assembly (44) on compensator studs at front of rear fireseal.

(4) Fasten compensator and seals to fireseal mount ring with self-locking nuts. Torque 36 to 40 lb.in.

(5) Connect heated line (1) to straight nipple (31). Torque 90 to 100 lb.in. and lockwire.

(6) Do pressure leak check (Ref. Adjustment/Test).

7. Cleaning/Painting

A. Cleaning

(1) Air Pressure Lines

CAUTION: DURING CLEANING, MAKE SURE THAT THE SOLVENT DOES NOT CONTACT THE RUBBER SECTIONS OF THE INSULATED AIR PRESSURE LINE ASSEMBLIES.

(a) Pressure flush the lines with petroleum solvent (PWC11-027) or (PWC11-031).

(b) Dry the interior and exterior of the lines using clean, dry compressed air.

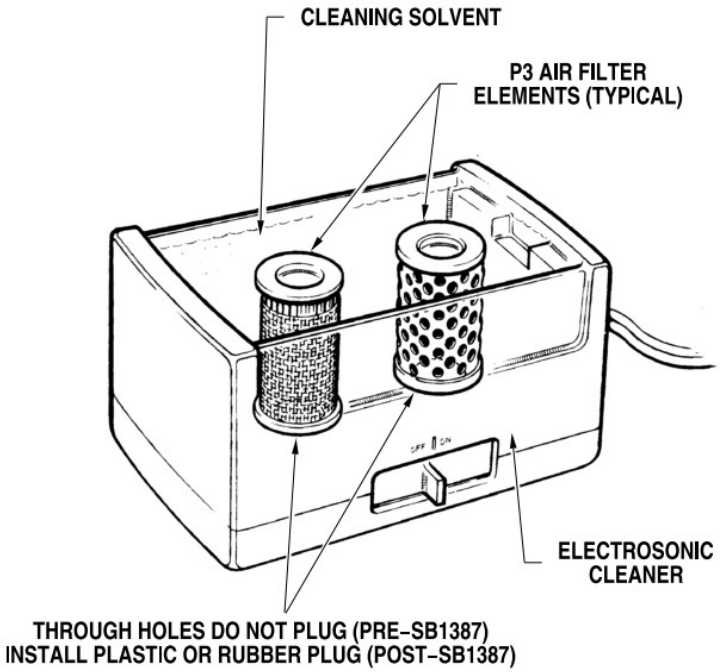
(c) Cap the ends of the lines to prevent the ingress of dirt, oil, or other contaminants until the lines are required for installation.

(2) Air Filter Element (Ref. Fig. 204)

NOTE: Inspect the filter element at the intervals recommended in Chapter 72-00-00, Table 601, and clean as required.

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Figure 204 Electrosonic Cleaner - P3 Air Filter Element



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(a) For Post-SB1387 Engines: Plug the opening at the end of the filter element with a suitable plastic or rubber plug.

(b) Place the P3 filter element vertically in the cleaning tank electrosonic cleaner with the open or plugged end of the element at the bottom position.

(c) Pour a sufficient quantity of cleaning solvent (PWC11-002) or (PWC11-014) into the tank to a level approximately 1/4 inch below the top.

(d) Operate the cleaner for ten minutes.

(e) Turn the element 180 degrees (with the open or plugged end of the element at the top).

(f) Operate the cleaner for five minutes.

(g) Turn the element to the original position in the tank (Ref. Step (b)).

(h) Operate the cleaner for five minutes.

NOTE: The cleaning duration should be adjusted to suit the condition of the filter element. However, for optimum results, the cleaning solvent should be replaced every 20 minutes.

(i) Wear clean, lint-free gloves, and remove the element from the tank. Remove the plug if installed.

(j) Dry the element with clean, dry compressed air (30 psi max.).

(k) Inspect the filter element for condition.

NOTE: If repeated cleaning using an electrosonic cleaner does not clean the elements to the required condition, the following procedures apply:

- A disposable element must be replaced with a new element.
- A permanent element may be shipped to an approved facility equipped with ultrasonic cleaning and pressure testing equipment and cleaned in accordance with the overhaul manual.

(l) Vibropeen cleaning date on end cap.

(m) Store cleaned filter elements in clean plastic bags until required for installation.

(3) Air Filter Housing (Post-SB1495):

(a) Clean cover (1, Fig. 202), spring (3), valve (4) and filter cover (6) with solvent (PWC11-027) or (PWC11-031) and non-metallic bristle brush.

(b) Dry with clean dry compressor air and/or clean with a lint-free cloth.

8. Inspection/Check

A. Inspection

(1) Insulated Tube Assemblies

(a) Examine flanged end fitting for elongated holes.

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(b) Examine tube assemblies (Ref. 70-00-00, STANDARD PRACTICES - INSPECTION, Inspection of Fuel, Oil and Air Tubes).

(c) Blend out damage that does not exceed specified limits. Replace tube assemblies damaged beyond limits.

(d) Examine the insulation on the pneumatic tube assemblies for cuts in the outer rubber sheath. Surface cuts up to three inches long may be repaired (Ref. Para. 9.).

(e) If surface damage exceeds specified limits above, but depth is superficial, sheathing may be repaired by replacement of damaged portion (Ref. Para. 9.).

(f) Examine the metal braid and electrical lead on heated tubes for cuts. Up to three broken wires per plait, or six broken wires per linear foot, are acceptable. Replace the tube assemblies where metal braid is chafed or worn through to the lead conductors.

(2) For Pre-SB1495 Engines P3 Filter Cover:

(a) Make sure vent hole in cover is clear of all obstructions (hole size 0.029 to 0.034 in. dia.).

(3) For Post-SB1495 Engines Air Filter Housing Drain Valve Assembly (Fig. 202)

(a) Filter Cover (6):

1 Check lockwire holes for damage (tearing, cracks).

2 Check threads for damage.

3 Check that contact area is free of nicks and/or scratches.

(b) Retaining Ring (5):

1 Check for distortion. Replace if distorted.

(c) Drain Valve (4):

1 Check that contact area is free of nicks and/or scratches.

2 Make sure drain holes are clear and unobstructed.

(d) Valve Spring (3):

1 Check for corrosion and/or damage. Reject if unsatisfactory.

(e) Drain Valve Adapter (1):

1 Check threads for damage.

2 Check that contact area is free of nicks and/or scratches.

3 Check flats and lockwire holes are free of damage.

4 Check retaining ring groove. Reject if damaged or worn.

B. Inspection of Air Filter Element

(1) Inspect air filter element at intervals recommended in Ref. 72-00-00, ENGINE -

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INSPECTION. Filter must be free of visible heavy contamination, and must not give unwanted restriction when blown through. If repeated electrosonic cleaning (Ref. Cleaning/Painting) fails to restore filter to an acceptable condition, replace disposable (non-stainless steel) elements. Non-disposable (stainless steel) elements may be returned to an approved overhaul facility for ultrasonic cleaning and flow check.

(2) Inspect the filter element for detachment of end cap or broken wires with a 10x magnifying glass. If any of these conditions are found, reject the air filter element.

C. Check

(1) Heated Tubes

(a) Using the turbine temperature indicating system test set, measure the resistance of the tube heating element and input lead. Resistance should be as noted in Table 201.

NOTE: 1. Resistance values are at 20°C (68°F).

NOTE: 2. For engine model applicability refer to IPC.

Table 201 Heated Tube Inspection

TUBE ASSEMBLY P/N	RESISTANCE VALUE (IN OHMS)	TOLERANCE (IN OHMS)
3011853	20	± 1.4
3014216	20	± 1.4
3021964	13	± 1.0
3028907	20	± 1.4
3030517 *	35	± 2.0
3030522 *	40	± 2.0
3031445 **	40	± 2.0
3031448 **	35	± 2.0
3031850	18	± 0.9
3011843	8	± 0.4
3027974	8	± 0.4
3032097	26	± 2.0
3032309	39	± 2.0
3045999-01	8	± 0.4
* Paralleled	18.5	± 1.5
** Paralleled	18.5	± 1.5

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TUBE ASSEMBLY P/N	RESISTANCE VALUE (IN OHMS)	TOLERANCE (IN OHMS)

9. Approved Repairs

A. Cuts in Insulated and Heated Air Pressure Lines

(1) Apply adhesive sealant (PWC08-002) or (PWC09-001) to the damaged area of the tube in accordance with the manufacturer's instructions.

NOTE: If the cut is near the maximum allowable, it may be necessary to bind the area lightly with plastic covered wire after an application of adhesive. Remove the wire after the adhesive has cured.

(2) Allow the adhesive sealant to cure at least two hours at room temperature. Full bond strength will be obtained in 72 hours.

(3) After curing, trim the excess adhesive if necessary.

B. Repair of Damaged Silicone Rubber Sheath

(1) Remove damaged portion of sheath, taking care not to penetrate to tape or insulation.

(2) Cut a length of silicone rubber sheath (PWC05-052) one inch longer than part to be covered.

NOTE: If a heated tube is damaged in the proximity of the electrical input lead, cut the repair sheath to allow for a minimum of $^{3}/_{8}$ inch overlap.

(3) Put sheath over tube assembly until overlap is equal at both ends.

NOTE: If damage is near pigtail lead on heated tube, make end of repair sheath flush with pigtail lead.

CAUTION: AVOID APPLYING EXCESS HEAT TO SHEATH AT EXISTING BOOT. FURTHER HEATING OF BOOT COULD HAVE ADVERSE EFFECT.

(4) Heat sheath all over with a hot air blower at 177 to 204°C (350 to 400°F). Sheath will immediately contract and should be smoothed out for a uniform fit.

(5) Trim sheath, as applicable, to 0.65 to 0.90 inch from end of ferrule. Seal ends of repair sheath with adhesive sealant (PWC08-002) or (PWC09-001).

(6) Allow sealant to cure for two hours at room temperature. Full bond strength will be obtained in 72 hours.

(7) After curing, trim excess sealant as necessary.

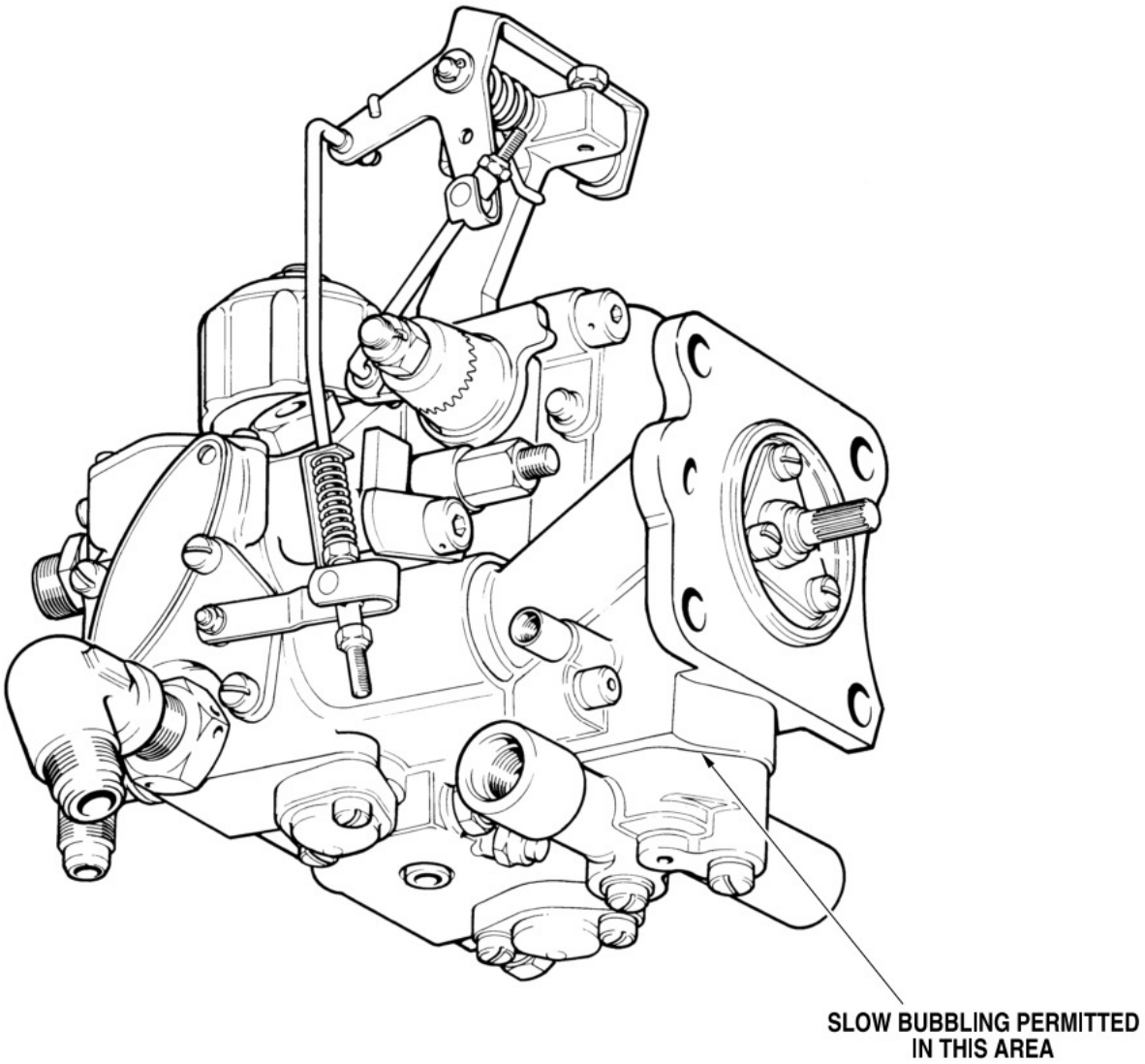
C. Replacement of Electrical Connector

(1) If replacement of the electrical connector is necessary, remove the tube wiring. Refer to Subpara. 6. for Removal/Installation of associated wiring.

10. Adjustment/Test

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Figure 205 Leak Check at Drive Body and Adapter Junction

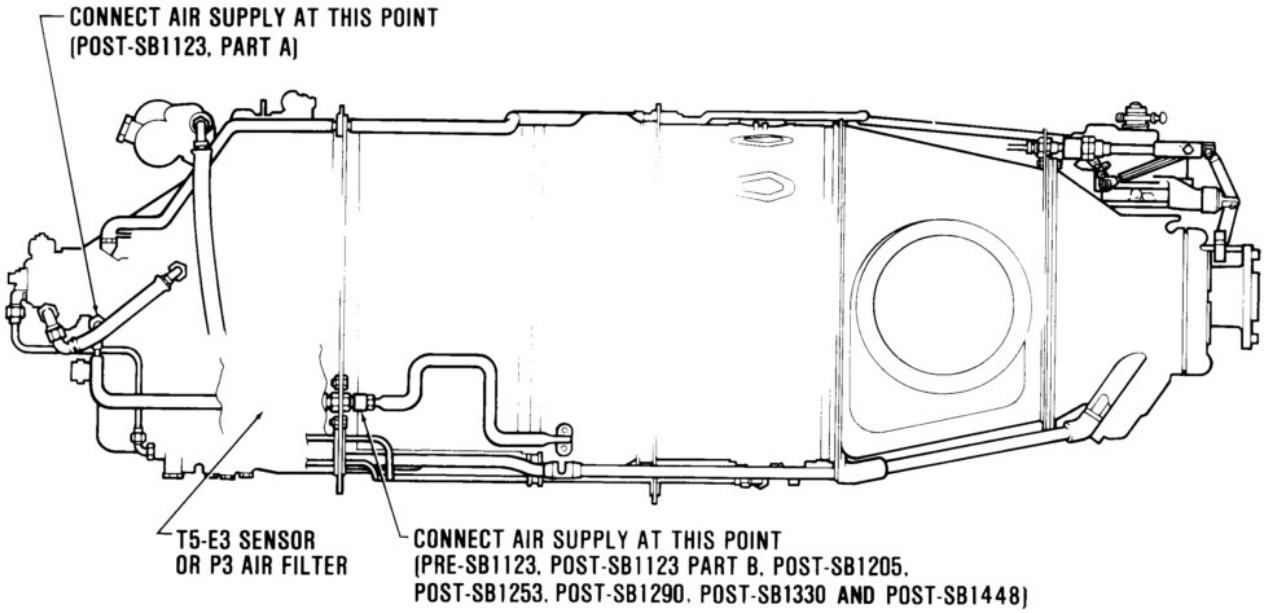


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A. System Test (Ref. Fig. 206)

Figure 206 Fuel Control Pneumatic System Pressure Test



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NOTE: After pneumatic lines are installed, and individual components within lines or components associated with lines are installed, pressure test the system.

(1) Dependent on engine configuration, disconnect compressor discharge air pressure line at appropriate connection.

(2) Connect a supply of clean, dry compressed air regulated to 80 ± 5 psig:

(a) For PT6A-27 and PT6A-28 Engines: Connect to elbow on compensator.

(b) For PT6A-27 and PT6A-28 Pre-SB1205 and PT6A-21 Engines: Connect to metered elbow on FCU.

(c) For PT6A-27 and PT6A-28 Post-SB1205 Engines: Connect to fitting at inlet of air filter.

(d) For PT6A-28 Post-SB1290 Engines: Connect to insulated center line at rear fireseal.

(e) For PT6A-21/-27/-28 Post-SB1330 Engines: Connect to insulated center line at rear fireseal.

(f) For PT6A-27 and PT6A-28 Post-SB1448 Engines: Connect to insulated center line at rear fireseal mount ring.

(3) Apply leak checking fluid (PWC05-007) to entire surface of all connecting points throughout pneumatic system. No leaks are permitted other than from one bleed hole underneath the fuel control unit.

NOTE: 1. For Pre-SB1495 engines, and one bleed hole in the P3 filter cover.

NOTE: 2. For Post-SB1495 engines, and minor leakage (Ref. NOTE 3) through the drain hole in the P3 filter drain adapter.

NOTE: 3. Excessive leakage at the P3 drain valve port (Post-SB1495) may result in unacceptable engine starting or acceleration characteristics.

NOTE: 4. Slow bubbling is permitted at junction of drive body and adapter (Ref. Fig. 205), provided that the engine accelerates normally (Ref. 71-00-00, Power Plant Adjustment/Test). Ship FCU to an approved facility for repair if leak is causing acceleration problems.

(4) If leaks are detected, affected parts must be inspected for defects before retorquing. Replace defective parts. If no defects are found, retorque connection and repeat pressure leak test.

(5) When no leaks are detected, wipe all leak checking fluid from connecting parts throughout system.

(6) Disconnect air supply from appropriate connection (Ref. Fig. 206).

(7) If front section of P3 line was disconnected at gas generator case, install new Post-SB1276 gasket (20, Fig. 201) flanged fitting and gasket to gas generator case using bolts, torque sequentially 15 to 20 lb.in., then continue torquing in same sequence 32 to 36 lb.in., and lockwire.

NOTE: 1. For Pre-SB1276 Engines: Do not lockwire bolts at flanged fitting end of line at

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this time.

NOTE: 2. For Post-SB1276 Engines: Do not loosen bolts to zero torque after applying initial torque.

(8) Connect coupling nut at rear end of tube (19 or 22), as applicable, tighten nut, torque 90 to 100 lb.in., and lockwire. Complete safety wiring of all connections throughout pneumatic system.

(9) Do engine start and ground run (Ref. 71-00-00).

(10) For Pre-SB1276 Engines: After ground run, retorque flange bolts at gas generator case boss and lockwire.

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